Texas Society of Professional Surveyors

Safety Handbook

Prepared by the Safety Committee of the
Texas Society of Professional Surveyors

TEXAS SOCIETY OF PROFESSIONAL SURVEYORS

SAFETY HANDBOOK

Prepared by the TSPS Safety Committee

Disclaimer

This Safety Handbook is provided as an educational resource to bring awareness to matters concerning safety in the surveying profession. The content within this handbook is intended to be informational in nature. This handbook is not to be substituted for medical advice from a qualified medical practitioner. Texas Society of Professional Surveyors makes no warranty as to the accuracy, relevance, or completeness of the information provided in this handbook.

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Introduction

Safety is an important aspect of surveying, especially in the field. There is inherent danger in just about all surveying field situations, including driving to and from the job safely. Knowledge of safe practices, proper use, and application of personal protective equipment (PPE), necessary care in selecting safe instrument setups, proper procedures, precautions in the use of hammering and cutting tools, as well as caution related to snakes, stinging insects and poisonous plants are imperative. According to a noted West Texas surveyor "everything in Texas either sticks, stings, or bites, or it wants to."

This handbook is not intended to be a complete volume on how to handle all aspects of safety or emergency situations, or to all problems that might occur in the field or in an office setting. The primary purpose is to provide relevant information and to prompt critical thinking and analysis of potentially unsafe situations and the hazards that may be encountered while working, both in the field and in the office, and to assist in knowing how to avoid or minimize the potential risks and dangers to prevent incidents and injuries. Use of this document is strictly voluntary. It is intended to serve as a guide with recommendations for safe practices.

Traffic control is an important aspect of surveying when measuring public roadways or working within a public right-of-way, when performing a survey adjoining a public roadway or even a residential street. When performing surveys adjacent to, along or within a city, County or State highway right of way (ROW), especially when near or on the roadway surface, traffic control becomes a more serious matter and should be implemented using the latest traffic control devises and set up plans recommended in the Texas Manual on Uniform Traffic Control Devices. Traffic control requirements, devices, plans and implementation shall be discussed with the jurisdiction having responsibility over the road and right-of-way. Strict adherence to all requirements is essential and all plans must be followed to ensure the safety of workers and the safe flow of traffic. It should also be noted that liability exists when the responsibility of controlling traffic is assumed.

I. Responsibilities

Employers have a responsibility, ethically and by law, to keep workers safe. Employees have an obligation to perform work safely and to protect the safety of other workers and others around them. The Texas Society of Professional Surveyors is committed to the education, safety and protection of its members and their protection from hazards that may occur by encouraging an active safety program and awareness of requirements, obligations, and good safe practices.

A. Employees

Employees shall perform work and conduct themselves in a manner that is safe according to policies and procedures of their employers and customary practices which do not inflict harm upon themselves, other workers, or the public. This includes use of equipment and tools according to manufacturer's instructions and manuals.

B. Supervisors, Administrators

Supervisors and administrators are responsible for seeing that employees are trained in safe practices for the conditions in which employees work and for the equipment and tools that they use in their work. Supervisors and administrators shall ensure that unsafe practices or conditions that exist or which are reported shall be addressed and that appropriate action is taken to remedy the situation or reduce the risk of danger or harm to a minimum. If additional tools, devices, or safety equipment is called for it shall be sought and put into use in the least amount of time possible.

C. Reporting Unsafe Conditions

Safety is the responsibility of every employee. All employees are responsible for reporting unsafe conditions. Open channels of communication between co-workers and up and down the chain of command is imperative and must be used.

If an unsafe condition exists, it should be remedied immediately before someone is hurt, injured or worse. If an unsafe condition is identified, a time out should be called by anyone and the situation should be discussed before work continues. If additional tools or devices are needed to remedy the situation, they shall be requested immediately.

D. Shall, Should, May

When these words are used in this document, the meanings are defined as:

Shall - a mandatory requirement. When used, it is mandatory that the requirements are met.

Should - an advisory condition. When used, it is considered as advisable or recommended though not mandatory.

May - a permissive condition. When used, there is no requirement.

II. General Guidelines

A. Situational Awareness

Situational Awareness is being aware of what is happening around you and understanding how information, events, and you own actions will impact your goals and objectives, both now and in the near future.

Each of us should, at all times, make it our habit to pay attention to the conditions surrounding of each situation in which we find ourselves. This especially applies to new, risky or dangerous situations often faced by a survey crew working in the field. It should also apply to normal, everyday situations such as driving and in office settings, failure to acknowledge the conditions around us and make adjustments to offset the dangers can result in injuries or even death.

B. Horseplay

Horseplay or other similar risky behavior should not be allowed. Horseplay is not a joking matter and should not occur. Serious injuries and/or unnecessary down time can result from horseplay. Employees may not be covered under workers' compensation if an injury results from horseplay.

C. Tailgate Meeting

An informal safety meeting, typically of a short duration and conducted onsite prior to work commencing. This meeting offers a supervisor, or survey crew chief opportunity to review important and relevant safety topics in an informal, conversational way. The primary purpose of the meeting is to identify and discuss hazards and conditions specific to the work tasks and location. This could include weather or other conditions that may change or vary during the duration of the work.

III. Allergic Reactions

Call 911 if a person has a severe allergic reaction, or has the symptoms listed below. Call 911 even if the person has an epi pen (a self-administered autoinjector device with epinephrine medication) and administers it to themselves.

- Wheezing or difficulty breathing.
- Tightness in the throat or a feeling that their airways are closing.
- Trouble speaking, hoarseness.
- Swollen lips, tongue, or throat.
- Nausea, vomiting or abdominal pain.
- Fast pulse or heartbeat.
- Anxiety, dizziness.
- Loss of consciousness
- Anaphylaxis - other symptoms of severe allergic reaction.

IV. Cardiopulmonary Resuscitation (CPR)

CPR is an emergency lifesaving procedure performed when the heart stops beating. Performing CPR could be necessary on the job, in the office, at home, or in any situation around other persons. All employees should be trained in performing CPR. CPR is best performed by trained and certified individuals. At a minimum, supervisors, survey crew chiefs or other lead workers should be trained and certified.

The two common versions of CPR are:

- For those trained and healthcare providers, CPR applying chest compressions and mouth-to-mouth breathing.
- For the general public or for those who witness someone suddenly collapse, using compression only or Hands Only CPR.

If CPR becomes necessary, the two steps are:

1. Call 911.
2. Using Hands-only CPR, push hard and fast in the center of the chest. Press near the center of the chest, 100 compressions per minute, about 2” deep.

V. Calling 911

When calling 911, provide the following information to the dispatcher:

1. Your name.
2. The phone number you are using.
3. The location and number of victims.
4. The victim’s approximate age, sex, and condition - is the person responsive? Breathing? Bleeding?
5. What happened to the victim and any special circumstances?
6. What is being done for the victim?

VI. Cables, Chains, Winches, and Towing

In a situation where there is a need to pull a stuck vehicle, or to tow a stranded vehicle, the first and safest option is to consider calling a tow truck, someone who is experienced in the use of cables or chains to pull or tow a vehicle. An improperly connected cable or chain or one that breaks under a load can cause severe injuries to anyone nearby and in the swing of the broken cable or chain.

The next safest option is to use a woven fiber tow strap. A woven tow strap designed for towing is made to stretch where a steel cable or chain has little to no stretch.

Before using a tow strap, inspect the strap for cuts, damage or tearing. Do not use if damaged beyond a safe condition. Discard and replace a damaged strap.

When handling a cable or chain, wear leather gloves.

Inspect all cables, chains, winch cables for kinks, frayed portions, damaged, stretched, or broken links. Do not use a cable or chain that is frayed or damaged as it can break

and cause severe injury to bystanders within the arc of a cable or chain that breaks under pressure.

Only hooks with safety latches shall be used on cables or chains.

Store cables, chains or tow straps in a protected place that will not allow damage to the device in between use.

VII. Confined Space

Confined space means a space that:

- Is large enough and so configured that an employee can enter and perform assigned work.
- Has limited or restricted means for entry or exit (for example, tanks, vaults, and pits are spaces that may have limited means of entry.)
- Is not designed for continuous employee occupancy.

The Occupation Safety and Health Administration (OSHA) uses the term "permit-required confined space" (permit space) to describe a confined space that has one or more of the following characteristics:

- Contains or has the potential to contain a hazardous atmosphere.
- Contains a material that has the potential to engulf an entrant.
- Has walls that converge inward or floors that slope downward and taper into a smaller area which could trap or asphyxiate an entrant.
- Contains any other recognized safety or health hazard, such as unguarded machinery, exposed live wires, or heat stress.

NEVER enter a confined space alone. If someone who entered a confined space goes down, DO NOT attempt to rescue the downed person without seeking additional help and by employing all necessary measures to ensure the safety of the rescue.

Before entering a confined space make certain that the conditions of the space have been assessed and analyzed; all requirements are known; required permits are secured and on hand; that all necessary safety equipment is on hand, in good working condition and utilized properly; and that all workers are briefed on the entry operation and the tasks to perform.

VIII. Driving

Driving defensively is important to being a safe driver. It is driving to save lives, time, and money despite conditions around you and the actions of others.

Every situation is different and calls for its own specific actions though there are five basic defensive driving tips that should always be employed to reduce the risk to yourself and others around you.

Look Far Ahead

Look as far ahead as you can see, also known as "aim high in steering". Plan ahead by looking ahead 15 seconds. In the city look to the end of the block, to the intersection ahead, and beyond if possible. On the highway, look ½ to one mile, to the next turn in the road or to the top of the next rise. Many accidents could be avoided with better situational awareness and by looking farther ahead.

Get The Big Picture

Maintain a four second following distance. Scan your mirrors, rear view and both side mirrors, often. Stay alert and avoid focusing on insignificant objects.

Scan and Take In As Much As Possible

Avoid focusing on objects too long, this diminishes peripheral vision. Keep your eyes moving. Eye activity stimulates the brain and resists effects of fatigue. Avoid distracted driving, inside your vehicle and in your thinking. Be alert for distracted drivers. Scan ahead and anticipate what could go wrong, prepare for the worst while taking in all that you can. Watch for stopping vehicles, children playing, and backing vehicles.

Have an Escape Plan, Leave Yourself An Out

Do not get blocked in by other vehicles traveling near you. Position yourself so that you leave space around you and have a way out if something goes wrong. Be a proactive driver and anticipate possible actions of other drivers. Maintain your space by adjusting- slowing, safely changing lanes, etc.

Make Sure They See You

Search for potential danger. Seek eye contact and make sure they see you. Use your signals, though not too soon or too late. Be sure that your signals are heeded. Do not assume other drivers or pedestrians see you.

IX. Emergency – First Aid

What to do in the event of an emergency:

- Recognize the emergency.
- Check the scene for safety; ensure there is no danger to you or others.
- Check the person and the scene. Never enter an unsafe scene.
- Call 911.
- Obtain consent and provide care, only to the extent you are trained.
- Have the person seek medical attention when needed.

If an accident occurs:

- A person who understands first aid should take charge.
- Stay calm.
- Keep the crowd away.
- If injuries occurred call 911. Wait for an ambulance or emergency response vehicle if possible. If not, take injured to an emergency clinic or

hospital emergency room if required. When calling 911 or emergency response provide the following information:

- Where the patient is and how to reach them.
- What you believe is wrong with the patient.
- What you are doing for the patient.
- In the meantime, do what you can to help the injured person.
- If the patient has been moved before the emergency response personnel arrive, leave someone at the scene of the accident to tell the team where to go.

Five most important things to do in case of injury:

- Stop severe bleeding
- Restore breathing
- Treatment for shock
- Proper transportation of injured
- Proper first aid treatment for wounds

X. Fire Extinguishers

Offices should have fire extinguishers located in readily accessible locations and at appropriate distances for use in the event of fire. Company vehicles, trucks, cars motorized off, road vehicles and boats with motors should all have a fire extinguisher mounted in or on each unit. If fire extinguishers are available for employee use, OSHA charges employers with the responsibility to educate employees on the practices and principles of using a fire extinguisher and the hazards associated with fighting small or developing fires. This training must occur annually, and new employees should be trained on these practices when hired.

When Using a fire extinguisher, follow these steps below:

- Sound the alarm, call 911 or the fire department.
- Use a fire extinguisher appropriate for the type of fire. Do not use water on an electrical fire.
- Discharge the fire extinguisher within its effective range following the P.A.S.S. technique.

P – PULL. Pull the pin making sure to break the tamper seal.

A – AIM. Aim low while pointing the extinguisher nozzle or hose at the base of the fire.

S – SQUEEZE. Squeeze the handle to release the contents of the extinguisher.

S – SWEEP. Sweep from side-to-side spraying at the base of the fire until it appears to be out. Watch for re-ignition of the fire. If it does re-ignite repeat Aim, Squeeze, Sweep.

If unsuccessful in extinguishing the fire, EVACUATE IMMEDIATELY.

XI. Hand Tools – Axe, Brush Blade, Hammer, Machete, Pick, and Shovel

Only use the right tool for the job. A wrench is not a hammer, a screwdriver is not a chisel or pry bar.

Wear appropriate PPE when using hand tools and cutting tools. Safety glasses, face protection, gloves and safety-toe boots are recommended.

Store tools in a safe place when not in use, especially when carried in a moving vehicle. Sharp edged tools should be stored with the edge protected. Example: a machete should be in a sheath when not in use or when being carried.

Maintain hand tools. Inspect tools for broken handles, loose hammer heads or other damage. Repair damaged tools properly. Replace if not repairable.

When sharpening or filing sharp edged tools, use extreme caution to prevent the file from slipping and causing injuries.

A. Cutting Tools - Axes, Brush Blades, and Machetes

- Keep cutting edge sharp and assure that the handle-head connection is tight.
- Wear eye protection. Leather gloves should be worn when using.
- Be sure others are clear before swinging. Have an area to swing that is clear of overhead branches, brush, and vines. A glancing blow of a cutting blade can strike a bystander and can cause severe injury.
- Ensure that your footing is stable.
- Protect the cutting edge with a sheath when carrying or not in use.

B. Hammers

The user should wear safety glasses, safety glasses with side shields or goggles. This applies even when simply driving a nail in pavement or a wood stake in the ground. A glancing blow to a nail turns it into a deadly bullet-like projectile. Wood stakes can splinter into flying projectiles. Iron rods or rebars can splinter sending metal shards flying. Wear gloves to protect hands from missed blows of the hammer, splinters, or flying metal shards.

C. Picks

Always wear eye protection. Make sure that bystanders are not near a worker swinging a pick to avoid flying debris or being struck by the pick.

D. Shovels

Use shovels for digging only. They are not pry bars. Keep handles, especially wooden handles in good shape and make sure that cracked handles are replaced or that the shovel is replaced.

XII. Hazardous Materials

A. Cement, Bagged Concrete Mix

Cement consists of the following ingredients: lime, silicone dioxide, aluminum oxide, magnesium oxide, iron oxide, calcium sulfate, sulfur trioxide, and alkaline.

Concrete consists of the same ingredients along with sand and aggregate.

Without proper protection, these chemicals pose hazards ranging from irritation, damage to eyes, skin, and the respiratory tract to anyone who handles bagged concrete or works with concrete mixture. It is recommended to use gloves, long sleeves, and appropriate respiratory protection when handling bags of the dry mix. When placing concrete for monuments, use rubber or vinyl gloves. If the dry or wet mix gets on skin, wash with generous amounts of soap and water. If clothing becomes soiled with cement or concrete, wash thoroughly as soon as possible.

B. Flammable Liquids - Gasoline, Diesel

Portable fuel containers should be red for gasoline, yellow for diesel.

Store gasoline and diesel fuels indoors, in approved storage containers with working safety caps and spouts, away from flames or potential heat sources - heaters, or water heaters.

Keep containers tightly sealed; handle carefully to avoid spills.

When transporting fuels, be sure to firmly tighten all caps on the container to avoid spillage. To prevent tipping or movement, secure the container upright (in the bed of a truck or attached to a trailer, with straps. Never transport fuel containers (either full or empty) in the passenger compartment of a vehicle.

Before filling, stop vehicle engines, generators, or other motors. Extinguish any open flames in the proximity.

Before touching the fuel container, ground yourself by touching metal on your vehicle.

ALWAYS place a fuel container, especially a plastic fuel container, on the pavement or ground to fill it. NEVER fill while it is in the bed of a vehicle. Static electricity buildup can cause a spark that can ignite the vapors. Allow a small air space for expansion of the fuel in a filled container. Once done, tighten all caps and lids.

Before fueling any fuel tank, always shut off a running engine- truck, car, ATV, UTV, generator or any other nearby engine.

No smoking while refueling.

C. Paint, Aerosol Cans

Always store aerosol spray paint containers, insect spray, or other pressurized containers out of direct sunlight and away from sources of open flames or heat. Extra stock can be stored indoors in an enclosed cabinet for greater safety.

XIII. Housekeeping

A. Field - Survey Crew

Maintaining good organization in a survey vehicle is good safety. That applies to traveling to and from work sites as well as when reaching for stored tools or supplies.

Keep equipment, tools, and supplies safely stored in compartments or toolboxes in the bed of truck, safely away from the passenger compartment. Cutting blades should be properly sheathed, both in transit and when carrying in the field.

Objects in the passenger compartment such as instrument cases, lunch bags or other daily items carried could become a flying object in a traffic incident. Always keep them low and secure.

B. Office Staff

Good housekeeping around an office is also a matter of safety. To prevent slips, trips, or falls:

- Keep flooring and carpet in good condition. Repair or replace damaged flooring. Repair loose, frayed, or torn carpet.
- Remove liquid spills promptly.
- Stack boxes in appropriate locations near walls or on adequate shelving. Do not store boxes on stairs, in frequently used hallways, traveled paths or on top of cubicle shelves or file cabinets.
- Do not block stairs, exit doors, or fire extinguishers.
- Place heavy items on the bottom of stacks, on lower shelves, or in bottom file drawers.
- Keep file drawers closed when not in use. Do not leave them open, as they can be a tripping hazard.
- Keep extension cords, power cords and cables bundled and out of walkways.
- Portable heaters should be kept a minimum of 3’ away from objects and not plugged in with multiple other electrical devices.
- To prevent fires or electrical shock, do not overload power outlets or power strips with an excessive number of devices.

XIV. Insects - Bites, Stings, and Allergic Reactions

(See Section XXI on Snakes, Spiders and Stinging Insects)

Texas has many insects and critters that sting or bite. Some persons are highly allergic to these stings and bites. Bee stings result in more deaths than snake bites. If a person is stung by a bee or wasp, bee sting swabs may offer relief. If a person knows they are allergic and they carry an epi-pen, use it. Or if they exhibit what could be an allergic reaction, take them to an emergency medical facility.

Persons who have allergic reactions to any of these should always be sure to inform their co-workers in case of a bad reaction.

Spiders also can bite and pose a danger to a bite victim.

XV. Off Road Vehicles - ATVs, UTVs, and Marsh Buggies

Off road vehicles (all-terrain vehicle) can save a surveyor a lot of walking and hard work packing equipment and supplies in and out of a job, as well as on the job. But they can be extremely dangerous when not used properly and safely. Most ATV injuries are because of misuse or operator error, or because the operator was not wearing proper personal protective equipment for riding such as a helmet, goggles, boots, gloves, long pants, and long sleeves. Other reasons for injuries include:

- Operating at excessive speeds resulting in loss of control and failure to detect hazards.
- Reckless driving and improper use of an ATV.
- Inadequate operator experience.
- Carrying passengers who altered weight balance, limited control, or restricted vision.
- Collisions with other vehicles.

The State of Texas has laws specifying the requirements for operation of an ATV on public property and on public road right-of-way. Some of these requirements are:

- Must have a state-approved safety course certificate, carried with the person/vehicle.
- Brakes, muffler system and U.S. Forest Service qualified spark arrestor, in good working condition.
- Working headlights and taillights.
- Must wear approved safety helmets and eye apparel.

A. 4-Wheelers and UTVs

- Attend a certified training course in how to operate these types of vehicles.
- Read and become familiar with the operator's manual before riding the first time.
- Wear an approved helmet designed to absorb blows and appropriate protective gear, long sleeved shirts, long pants, boots, and gloves.
- Do not carry passengers on a single rider seat ATV, and no more in a multi-seat UTV than the number of seats and seat belts provided.
- Do not carry loads or attachments until you have been trained in driving with attachments or loads.
- Maintain safe speed for the terrain and conditions in which you are riding.
- Do not operate while under the influence of medications, alcohol, or drugs.
- Be alert for potential hazards, unexpected wildlife.
- Should have an approved fire extinguisher of a size appropriate for the vehicle.
- Shut off the engine when refueling.

B. Boats

See Section XXVII on Water Operations.

C. Marsh Buggies

A marsh buggy is a specialized motor vehicle for use in marshes and swamps, having wide treads or large rubber flotation tires with cleats. If one is required for survey work to be performed, make certain that the entire survey crew and especially the operator has been thoroughly trained in the operation and use of the specific unit. Improper use can subject the operator and passengers, as well as the equipment carried, to such danger as trapping under water and death by drowning. Certain designs may be prone to tipping over when floating in water deep enough so that the tires do not touch bottom.

XVI. Poisonous Plants

Poisonous plants, poison ivy, poison oak or poison sumac are present over much of Texas, and other states. Learning to identify and avoid them is the first and foremost way to prevent exposure to these plants. Application of poison ivy protective barrier cream is another form of protection that is readily available. Wear long sleeves, gloves and long pants also offers protection.

Exercise caution when walking thorough, reaching into, or cutting vines and leaves of any poisonous plants as simply brushing leaves or cut vines can allow particles**,** oil**,** or sap to contact skin or clothing. Transfer of this sap to other objects or persons is also possible. If clothing has come in contact with poisonous plant sap, make sure to wash this clothing as soon as possible. Wash contaminated clothing separately, do not mix with other clothing as the sap can be transferred to other clothing.

Use caution with cutting tools or other objects such as a prism or GPS rod that has been in contact with vines or leaves. The sap or oil can be transferred and continue to contaminate other persons or objects. Do not burn as the smoke or dust can infect eyes and the respiratory tract.

If contact with skin is suspected, a post exposure cream is recommended along with washing the exposed skin as soon as possible.

Exposure usually results in an itchy rash that develops in hours up to days after contact. Skin becomes red and itchy, with blisters usually developing. Swelling and fever could develop if left unattended. Over the counter treatments may be effective, but medical attention may be required.

Repeated exposure usually causes more severe reaction.

Poison Ivy

Can be found as a vine growing up a tree or utility pole or as plants/shrubs. Leaves may vary slightly in shape though all will have three leaflets, usually one with three points and one similar to a hand wearing a mitten. Leaf texture is waxy on top and dark green. The underside may be light green with fine hairs. In the fall leaves change to orange or red.

Greatest hazard is in spring and summer when the sap is more abundant.

Plants may have white with green tint flowers and pea sized fruit in clusters colored white or ivory with a green tint.

Poison Oak

May appear similar to poison ivy though its three leaflets are deeply toothed with rounded tips or lobed.

Found in the west usually in places, wooded slopes, or thickets; in the east in sandy soil, in pine or oak-pine woods.

Poison Sumac

May grow as a small tree or tall shrub. Leaves alternate with seven to 11 leaflets in pairs, with an additional single leaf at the end of the midrib. The plant may have yellow-greenish flowers with whiteish green fruit hanging in loose clusters. It is found growing in standing waters- swamps, bogs, river bottoms.

XVII. Personal Protective Equipment (PPE)

Personal Protective Equipment is always recommended when working where potential hazards may exist. PPE shall be worn or used when in a work area where it is required. It is the supervisor's and the crew chief's responsibility to ensure that PPE is available to all workers, that all workers have the proper PPE in good condition and that it is worn properly and as designed so that it offers the maximum intended protection. PPE should be kept clean and maintained in serviceable condition. When no longer serviceable (discolored or excess wear), it should not be used and should be replaced.

A. Eyewear

Safety eyewear (glasses) are for general purpose eye protection and should be worn where flying particles are present, when on or near roadway surfaces or overpasses, and in situations where brush or flying objects could impact a person's eyes.

All safety eyewear shall conform to the latest The American National Standard for High-Visibility Safety Apparel (ANSI/ISEA) standard (currently Z87.1). A person requiring prescription eyewear should use goggles, a face shield or they may wear prescription safety glasses conforming to this same standard.

Goggles are recommended in dusty conditions or when handling items such as bag concrete or cement.

B. Footwear

Always wear footwear appropriate for the conditions. High top boots are always recommended to offer the wearer ankle support as well as protection.

If safety toe boots are required by a client on jobs such as construction, industrial sites, working in a highway Right-of-Way (ROW), or any other location, field personnel shall wear the required footwear prescribed.

Footwear with a safety toe cap must conform to current requirements of American Standard for Testing Materials (ASTM) F-2412 and/or F-2413 for safety footwear. These standards include withstanding a 75-pound compression and impact as well as electrical hazards.

C. Gloves

Gloves should be worn when handling hazardous materials (concrete, cement, gasoline, chemicals), when hands are exposed to impact/pinching, cuts and abrasions, vibration conditions, extreme heat or cold, or electrical hazards. Leather gloves, rubber gloves, insulated gloves or other types may be appropriate depending on conditions.

D. Headgear

Safety hard hats should be worn when:

- Required by the client.
- Performing any work or task where there is danger of head injury from impact, falling debris, flying objects or electrical shock.
- Visiting or working in a public road ROW.
- Required by an employer, a client or jobsite rules.

E. Hearing Protection

Appropriate hearing protection in the form of ear plugs or earmuffs should be worn when working in an environment such as a construction zone or with a power tool such as a chain saw or gasoline powered drill. When the noise level exceeds an 85 decibel (db) level hearing damage may occur. Noise at a level at or above 90 decibels is considered as damaging, even if momentary or for short durations. Also

consider that many power tools generate noise at or below 85 db only when used at arms-length, but when held closer to the head/ears the noise level is greater.

F. Respiratory Protection

Respiratory protection is designed to protect workers against harmful dusts, fogs, smokes, mists, gasses, vapors, and sprays which may cause harm, impairment, or even death. These devices can range from a simple dust mask to reduce dust or allergens to a self-contained breathing apparatus (SCBA) when in a hazardous atmosphere. Consult a professional to make the appropriate determination.

G. Safety Vests, and Shirts

Survey field personnel should wear highly visible, bright colored shirts when working in the field. These types of shirts help make workers more obvious to the public. More importantly, they make field personnel more visible to drivers. Bright colors are not a substitute for safety vests or safety shirts as they are required when working on a public road, street, highway, or right-of-way.

An approved safety vest or safety shirt shall be worn when working on or near the pavement or within the right-of-way of a public road/street/highway. Consult the Texas Manual of Uniform Traffic Control Devices (TMUTCD) for additional information. The minimum standard for a safety vest or safety shirt for this application is ANSI 107, Class II (latest standard) which includes a minimum of 775 sq. inches of background material and 210 sq. inches of reflective materials.

ANSI 107 Class III safety vests or safety shirts provide a minimum of 1240 sq. inches of background material and 310 inches of reflective material. This class meets the minimum standard of the Texas Department of Transportation (TxDOT) for performing work within a state highway right-of-way.

Higher traffic speeds, weather conditions, limited visibility, night work, other adverse conditions or employer/client requirements may call for a higher class of safety wear or additional safety garments to meet minimum standards.

XVIII. Power Tools – Gasoline

Gasoline powered tools can be dangerous and cause serious injuries if not used properly. Operators of power tools should become familiar with the tool operation before using it by reading the manufacturer’s operation manual and become familiar with the safety features as well as the dangers and “pinch points” of each tool.

A. Auger

Before using an auger:

- Follow the instructions in the manufacturer's operating and preventive maintenance manual. Check controls, and all bolts and handles to ensure that they are functioning properly and that they are adjusted according to the manufacturer's instructions.

- Conduct daily pre-task meetings to ensure that all employees are aware of the correct procedures to prevent an unwanted incident and any hazards associated with the job task.
- Look for obstacles that may need to be removed. Hand digging may verify the presence or absence of underground material, including utilities.
- Remain a safe distance (a minimum of 10 feet) from the auger. Except for the operator, employees should not be near the auger when it is operating.
- Employees using hand tools should not move or remove spoil-pile while the auger is operating.
- Do not modify the operator's station or disable safety controls beyond manufacturer's recommendations.

B. Chainsaws

1. Prior to Starting a Chainsaw
    - Check controls, chain tension, and all bolts and handles to ensure that they are functioning properly and that they are adjusted according to the manufacturer's instructions.
    - Make sure that the chain is always sharp, and the lubrication reservoir is full.
    - Start the saw on the ground or on another firm support. Drop starting is never allowed.
    - Start the saw at least 10 feet from the fueling area, with the chain's brake engaged.
2. Chainsaw Safety
    - Clear away dirt, debris, small tree limbs and rocks from the saw's chain path. Look for nails, spikes, or other metal in the tree before cutting.
    - Shut off the saw or engage its chain brake when carrying the saw on rough or uneven terrain.
    - Keep your hands on the saw's handles and maintain secure footing while operating the saw.
    - Proper personal protective equipment must be worn when operating the saw, which includes hand, foot, leg, eye, face, hearing, and head protection.
    - Do not wear loose-fitting clothing.
    - Be careful that the trunk or tree limbs will not bind against the saw.
    - Watch for branches under tension, they may spring out when cut.
    - Gasoline-powered chain saws must be equipped with a protective device that minimizes chain saw kickback.
    - Be cautious of saw kick-back. To avoid kick-back, do not saw with the tip. If equipped, keep tip guard in place.
    - Do not operate when fatigued. Make sure to take sufficient rest breaks.

C. Fueling a Gasoline Powered Tool

- Use approved containers for transporting fuel to the saw.

- Dispense fuel at least 10 feet away from any sources of ignition when performing construction activities. No smoking during fueling.
- Use a funnel or a flexible hose when pouring fuel into the saw.
- Never attempt to fuel a running or HOT motor.

XIX. Railroads and Railroad Rights-of-Ways

A. Basic Railroad Rules

Entry into a railroad right of way is not only unsafe it is unlawful unless prior arrangements are made with the respective railroad or railroad operator and all requirements of the railroad are met.

Prior to performing any work within or on a railroad right-of way or within 25 feet of a highway-railroad grade crossing the railroad must be contacted.

Do not set traverse or control points, monuments, stakes or leave flagging in a railroad right-of-way. Do not place paint, especially red paint. In the railroad world, red indicates immediate danger.

Always expect movement of trains or railroad equipment on any track, in any direction, at any time.

Additional care should be exercised when multiple tracks exist; trains on separate tracks may run in either direction.

Avoid fouling the track(s). Do not set up an instrument, park a vehicle or work within 25 feet from the outside of an outside rail. Fouling space is a total of 55 feet wide. 5 feet for track width (4 feet 8 ½ inches standard tack gage) plus 25 feet plus 25 feet from each track = 55 feet total.

Each railroad crossing is assigned a specific crossing number, usually posted on a blue sign. A 1-800 telephone number should also be posted at each railroad crossing of a public road. This number can be used in an emergency to contact the dispatcher of the railroad operator.

B. Right of Entry Requirements

Persons who require entry to work anywhere within a railroad right-of-way must be pre-certified and have an E-rail certification.

Work within 25 feet of the nearest rail will also require having a Roadway Worker-In-Charge (RWIC) to be present onsite to ensure that no one or their equipment is within the Fouling space and to monitor train traffic.

Railroad certification can be a lengthy process and there is a cost. Allow sufficient time for this process. Failure to comply with training requirements or right of entry protocol can result in fines imposed by the railroad company that has ownership or responsibility for the right-of-way.

XX. Surveying Field Safety

A. Drinking Water

If a water cooler is used for drinking water, do not place other items - can or bottle drinks, food, etc. in the cooler. For other items, use a separate cooler.

Water Containers - Use only clean containers which have been designated and used only for drinking water. Fill only with fresh water (cases of hepatitis have been traced to dirty and improperly maintained water containers). Do not use for cooling or storing canned or packaged beverages, food, etc. For other items carried, use a separate cooler that is used only for this purpose.

Cups - Use disposable cups or personal, reusable cups only.

Ice - If ice is added to the water, it shall be transported to the water container in a sanitary wrapping or clean container.

Keep the inside and spout of a water cooler clean. Sanitize with appropriate sanitizer solution often.

B. Driving Nails, Spikes, Property Corners, Stakes, etc.

Safety eyewear, PPE, should always be worn by all persons when driving a nail, rebar, spike, any metal objects, or when using a chisel, including other workers in the immediate area. The object being driven, especially hardened nails driven into pavement can become a bullet like projectile that could strike the person or a bystander.

Any metal object can break when struck by a hammer and produce a metal shard that can become a bullet like projectile that could strike the person or a bystander.

A chisel can produce flying chips of hard materials- rock or concrete, that can injure the person or bystanders.

Rebar driver devices and chisel holders that protect hands should be used.

C. Fences

Crossing a fence by climbing is always discouraged. This is especially applicable to barbed wire fences and chain link fences with protruding wire tops. Use a gate when possible as slipping can cause injuries. If climbing is necessary, use extreme caution and get assistance from a crew member. Fence climbing devices are available and increase safety and reduce possible damage to the fence.

Do not carry survey instruments or other tools or objects when climbing a fence. Hand them to a crew member or place them near the fence, cross the fence first, then reach for the instrument or objects.

D. Lasers- Electronic distance measuring devices.

- Never look directly into a laser device, from any distance.
- The instrument person should never point a laser device directly at a person.
- Know and observe all manufacturer's instructions and precautions.

E. One - Person Survey Crew

A one-person survey crew offers the advantage of economy and efficiency, though a one-person survey crew can be risky and almost certainly will expose the person who does not have a partner/spotter to the risks of unforeseen dangers and to the risk of theft of expensive robotic survey instruments being left unattended and vulnerable if the individual works too far from the instrument. If working alone:

- Be alert, treat the situation just as walking and texting, talking on a cell phone or distracted driving.
- Have good situational awareness.
    - Do not pay more attention to your tasks than to your surroundings.
- Do not work alone near the travel lanes or on pavement in traffic.

F. Paint and Pressurized Containers

Pressured cans such as spray paint or insect repellant pose of danger of explosion due to excessive heat or from puncture, as well as the danger of the contents being sprayed upon someone nearby. They should always be stored in a place where they out of direct sun light, away from the passenger compartment and where they are reasonably secured during travel. In the bed of a survey truck or the trunk of a passenger vehicle is recommended.

G. Powerlines

Contacting an energized power line can be dangerous, even deadly. Always be aware of powerlines, both overhead and buried.

Be aware of your surroundings before beginning work as well as while working where the possibility of contacting or getting near a powerline exists.

Keep a minimum of 10 feet away (in all directions) from any overhead powerline. Higher voltage lines require even more separation distance. When uncertain of a power line's voltage, stay 20 feet away for voltages up to 350 Kilovolt (kV) and 50 feet away for voltages greater than 350kV.

Aluminum, carbon fiber, steel rods, poles or tapes will conduct electricity.

Even non-metallic objects such as fiberglass rods, tapes, or wooden level rods can sometimes conduct, especially when wet.

XXI. Snakes, Spiders, Stinging Insects

A. Venomous Snakes

Texas has four types of venomous snakes: rattlesnakes, copperheads, water moccasins and coral snakes. Any snake may bite, especially when threatened, stepped on, cornered or surprised. Any snake bite may result in severe injury or death, even if no venom is injected or if the snake is non-venomous.

Precautions to prevent snake bites vary depending on the type of snake, though some are universal. The best precaution is to be alert, walk carefully and avoid snakes.

Wear boots and long pants. Wear snake leg protectors or snake boots when in areas of greater possibility of snake encounters.

Scan the area constantly and watch where you step. Do not step over logs or into brush without looking first. Be especially observant for snakes waiting for small game near trails or paths.

Do not reach into brush, under logs or into crevices without gloves, and without checking or probing with a stick first. Be especially watchful when moving or lifting objects off the ground or moving logs or brush piles. Snakes may be found beneath or within these objects.

A coiled snake can strike approximately one third of its own length.

Some snakes can be found in trees.

Snake Bites:

The best action is to avoid snakes, if possible. However, if bitten by a venomous snake, assume that the snake injected venom:

- For a rattlesnake bite hold the hand or limb at the victim's heart level.
- For cotton mouth and copperhead - keep the affected hand or limb above the victim's head.
- For a coral snake bite, try to stay calm and move body so bite is below victim's heart.

If a snake bite occurs:

- Keep the victim calm.
- Have them lie down. Avoid using the affected limb if possible. Do not walk on the affected leg unless you are alone, without help.
- Clean the bite with soap and water.
- Remove tight-fitting clothing, rings, watches, or jewelry as swelling may occur.
- Try to remember the type of snake, colors, and size. DO NOT try to catch it.
- Call 911 or other emergency services. Alert them of a snake bite victim.
- If calling is unavailable or emergency services cannot reach the victim carry the victim if possible. Otherwise, assist the victim to walk to the vehicle.
- Transport the victim to an emergency room or medical care facility.

Rattlesnake

There are 10 species of rattlesnakes in Texas that vary in color and size. The most common are the Western Diamondback and the Timber Rattler.

Listen for the sound of rattles.

The majority of snake bites in the U.S. are from rattlesnakes.

They may be aggressive if surprised, disturbed, cornered, or stepped upon.

Try not to walk near rocks, rock walls or brush piles where they may frequent.

Copperhead

A copperhead seldom bites unless disturbed, surprised, stepped upon or cornered.

Cotton Mouth *(Commonly known as Water Moccasin)*

The world's only semi-aquatic viper. Heavy bodied.

Found in swamps, coastal marshes, sluggish waterways, rivers, streams, and ponds. Sometimes found in swimming pools.

Seldom bites unless surprised, stepped upon, cornered or disturbed.

Other similar looking non-venomous water snakes are often more aggressive.

Coral Snake

Red and yellow kill a fellow, red and black venom lack.

Coral snakes are usually not aggressive unless disturbed or cornered. Coral snakes have no fangs, they must chew or bite and hold to inject their venom. They typically bite in soft tissue such as between fingers.

B. Spiders

Texas has two groups of venomous spiders and almost 900 other species. Non-venomous spiders can and do bite. Their bite may be painful, but it is not typically deadly.

The best protection from spiders is prevention.

Venomous spiders include the Black Widow and the Brown recluse.

If bitten by a venomous spider seek medical treatment as soon as possible.

Black Widow

Prefers cavities outdoors, found in portable toilets, storage sheds and undisturbed places. It is a shiny black color with a red hourglass pattern on the underside of its abdomen. Usually 1 ½" long. Only the female is dangerous to humans; the bite may not be felt or may feel like the prick of a pin. The bite site may have slight, local swelling with two faint red spots and redness. Pain may increase in 1 to 3 hours and continue for up to 48 hours. Pain may localize in the victim's back and abdomen, also in the muscles and soles of feet. Nausea, profuse sweating, tremors, labored breathing, difficulty in speech and vomiting may occur. Although black widow bites are uncommon, if medical attention is not sought immediately, cold, clammy skin, unconsciousness, convulsions, infection of the bite and possible long-term complications may develop. Death is rare but can occur.

Brown Recluse

Named for its shy nature. Most active at night, tends to hide during the day in and around buildings in warm, dry places. Size is usually ¼" to ¾" long. Often called the fiddleback spider due to the violin like design on its back, between their eyes and abdomen. Venom of a brown recluse can cause death and decay of tissue surrounding the site of a bite.

C. Stinging Insects

Stinging insects are found around homes, buildings, in fields and in the woods. The first defense is to avoid contact with them, if possible, or limit exposure. Long sleeves and gloves are recommended as protection.

Scorpions

Have four pairs of legs, a segmented tail with a poison gland and stinger on the end, and two pinchers. Commonly found under rocks, boards, and other litter. Most active at night. The sting is similar to a bee sting, causing pain and local swelling. Rarely is it serious though unless allergic reaction is experienced. This calls for medical attention.

Bees

There are several types of bees in Texas. They are found in colonies or hives, usually near flowers, gardens, fields, and forests. Bees differ from a wasp in that they are furry looking, thick bodied and short and rounded. Bees usually sting in self-defense or when threatened. A bee stinger is barbed and is left in the victim when they sting. The bee dies afterwards. The most prevalent is the Honeybee, identified by the orange and black stripes.

Bumblebees are larger than honeybees, have a black body and are covered with thick yellow and black hair. Bumblebees can sting repeatedly.

Other less common bees may be shaped slightly different and can be bright green or dull dark colors.

How to Avoid - If a bee flies nearby, slowly raise your arm to protect your face and stand still or move slowly to escape.

Treatment - If stung and the stinger is still embedded, do not pull it out as it can release more venom. Use a credit card or stiff cardboard and scrape it out. A bee sting may cause swelling, breathing difficulties, severe drop in blood pressure and shock. Allergic reactions are serious and medical attention may be required or call 911.

Africanized Bees

Africanized bees look like regular honeybees though microscopically smaller. Venom is the same as a honeybee. They are significantly more aggressive, even when unprovoked, and tend to swarm their target with hundreds of stings.

If attacked:

- Alert others.
- Cover your head and face, do not block your vision.
- Exit - run and get into vehicle or building. Do not waste time trying to find the colony.

Hornets/Wasps

Hornets are related to the wasp family. They have rounded abdomens and a wider body.

Wasps are long, slender, sleek insects, often colorful, have antenna, a pinched waist, and a stinger. There are several varieties and are usually aggressive, especially when provoked. They can sting repeatedly and do so often. They do this to obtain food. As with a bee sting, swelling can occur as well as possible allergic reaction.

Baldfaced Hornet

Wasp

Yellow Jacket Wasp

Treatment - Seek medical attention if swelling or other symptoms appear serious.

Treatment of Bites and Stings:

- Remove a bee stinger with tweezers or by carefully scraping across it with a credit card or similar object.
- Wash spider bites with soap and water. Elevate and ice the area if possible.
- Apply antibiotic ointment.
- If an allergic reaction is experienced, seek medical attention immediately or call 911.
- May take an antihistamine as soon as possible.
- Use an epi pen if the victim is allergic and carries one.

D. Ticks and Chiggers

Ticks are usually found in wooded areas or high grasses where livestock or wildlife are known to roam, especially during spring and summer months.

To reduce exposure to or avoid ticks:

- Avoid tick habitat - brushy overgrown areas, grassy areas, or wooded areas.
- Wear light colored clothing to see ticks more easily.
- Wear long sleeved shirts, high top boots and keep pants tight around the ankles or tucked into boots.
- Use insect/bug repellant stated to be effective in repelling ticks. Spray clothing around ankles, at your waist area and sleeve cuffs. Spray on skin if the repellant is safe for direct contact.
- After working in tick habitat, check your clothing first for crawling ticks. Then check your body where ticks are known to locate such as behind your knees, between fingers and toes, underarms, behind ears, on the neck and wherever there is hair.
- Shower well and again look for ticks.

- Wash clothing in hot water, dry on high temperature.

Ticks carry Lyme disease, a bacterium typically carried by mice and other small rodents. Lyme disease is transferred by a tick biting an infected animal and then biting a human. If you feel sick or have a rash after a tick bite, seek medical treatment immediately.

Tick Removal - the primary goal is to remove the tick as quickly as possible. Other methods of using oil, petroleum jelly, heat, etc. are not reliable.

- Using fine tip tweezers, grasp the tick as close to the victim's skin as possible.
- Pull upward with even, steady pressure. Do not twist or jerk the tick, this can cause mouth parts to break off, remaining in the skin.
- Once the tick is removed, clean the bite area, and wash your hands with soap and water or rubbing alcohol.
- Do not touch the tick with bare hands, do not crush it with fingernails.
- Place the tick in a sealed bag, in alcohol or flush down the toilet.
- Be alert for fever or rash, seek medical treatment immediately and inform the medical professional about your tick bite - when and where.

XXII. Temperature Extremes – Cold, Heat, and Sun Exposure

A. Cold

Cold Stress - A condition faced by workers who are exposed to extreme cold which can lead to serious health problems. It results from the body losing heat faster than it can be produced. A body temperature that is too low affects the victims brain making them unable to think clearly or move well.

Hypothermia - is an abnormally low body temperature. Symptoms can vary depending on how long a person has been exposed to cold temperatures.

Symptoms:

- Shivering
- Fatigue
- Loss of Coordination
- Confusion and disorientation

Late Occurring Symptoms:

- No shivering
- Blue shin
- Dilated pupils
- Slowed pulse and breathing
- Loss of consciousness

Treatment:

- Alert the supervisor and request medical assistance.
- Move the victim into a warm environment.
- Remove wet clothing.
- Warm the center of their body first - chest, neck, head, and groin with loose, dry layers of blankets, towels, or similar items.

- Warm beverages may help increase the body temperature but do not give alcoholic beverages. Do not give beverages to an unconscious person.
- Keep the victim dry and wrapped in warm blankets.
- If the victim has no pulse, begin CPR.

1. Frostbite is an injury to the body caused by freezing. It causes a loss of feeling and color in the affected areas: nose, ears, cheeks, chin, fingers, or toes. Frostbite can cause permanent damage to body tissue, including amputation. The risk of frostbite is increased for improperly clothed workers in extremely cold temperatures.

   Symptoms:
   - Reduced blood flow to hands and feet
   - Fingers and/or toes can freeze
   - Numbness
   - Tingling or stinging
   - Aching
   - Blush, pail, or waxy skin

   Treatment:
   - Move a frostbite victim into a warm enclosure.
   - Unless necessary, do not walk on frostbitten feet or toes, this increases damage to them.
   - Immerse affected area in warm water. Do not use hot water.
   - Warm fingers by placing in the heat of an underarm.
   - Do not rub or massage frostbitten area as this may cause more damage.
   - Use extreme caution when warming near a heater, stove, fireplace, etc. Numbed areas can be easily burned.

B. Heat Illness - Sunburn, Heat Cramps, Heat Exhaustion, and Heat Stroke

Heat illness can strike anyone. It is caused by prolonged exposure to hot temperatures and/or humidity higher than average, especially in the summer. Heat related illness are avoidable by taking proper precautions such as: take hydration breaks as needed; wearing light colored, loose-fitting and/or cooling clothing that covers as much of the body as possible; long sleeve shirts, using sunscreen; drinking plenty of fluids, alternating water, and sports drink (not sugar filled, caffeinated, or carbonated drinks); wearing wide brim hats or using hard hat shades or a sun brim; avoid heavy meals before working in the heat. Conditioning is imperative when first beginning work in heat/sun by working shorter hours and building up to a full day; stop work when necessary, to prevent conditions from worsening into cramps, heat exhaustion or heat stroke.

1. Sunburn

   Symptoms: Painful, red, warm skin. Severe sunburn produces blisters.

   Treatment: Stay out of the sun until sunburn heals; put cool clothes on sunburned areas or take cool bath; apply moisturizing lotion if blisters develop do not touch.

Prevention: Wear long sleeve, loose fitting, light colored, cooling shirts, and hat with wide brim. Use sunscreen of at least sun protection factor (SPF) of 30. Apply sunscreen according to product directions, and often, especially after getting wet or washing.

2. Heat Cramps

   Heat cramps can be avoided by stopping the activity, moving to a cool place. Drink water or a sports drink. Wait for cramps to go away before resuming the activity.

   Symptoms:
   - Heavy sweating during intense work or exercise
   - Muscle pain or spasms

   Treatment: stop activity and move to a cool/shady place, drink water or sports drink before resuming the activity.

3. Heat Exhaustion

   Occurs when the body is not able to sweat enough to cool itself. Symptoms of heat exhaustion include:
   - Heavy sweating, cool, pale clammy skin
   - Dizziness, weakness, nausea, headache, or vomiting
   - Blurry vision
   - Body temperature of 101° F
   - Feeling hot and thirsty
   - Having difficulty speaking

   Treatment: A victim of heat exhaustion should be moved to a cool place, loosen clothing, sip ample amounts of water, place cool, wet clothes on body. Seek medical help immediately if vomiting occurs, symptoms worsen or if symptoms last longer than one hour.

4. Heat Stroke (Sun Stroke)

   Results from untreated heat exhaustion. Includes the following symptoms:

   - No sweating
   - Unaware of heat and thirst
   - High body temperature 103° F or higher
   - Hot, red, dry, or damp skin
   - Fast, strong pulse
   - Headache, nausea, dizziness
   - Confusion or delirium
   - Loss of consciousness
   - Seizure

   For a victim of heat stroke, call 911 or take the person to a medical facility immediately, this condition is life threatening. Move to a cooler place. Do not give the person anything to drink. Cool by placing wet towels around the person's, on arms. Watch the person carefully until medical attention is received.

C. Hydration

If you are thirsty, drink. Do not wait until you are thirsty to drink, especially when working in warm to hot weather. Do not forget to drink in cold temperatures. Your body needs to be properly hydrated to function correctly.

To determine how much water a person needs one of two methods can be considered. Method one: drink eight ounces (oz) of water eight times per day. Method two: divide your weight by two, then drink that many ounces of water per day. A 180-pound person should drink 90 oz of water daily. With either method, drink more when performing physical activity causing sweating, or while working in heat and humidity.

If a water cooler is used for drinking water, do not place other items - can or bottle drinks, food, etc. in the cooler. For other items, use a separate cooler.

Keep the inside and spout of a water cooler clean. Sanitize with appropriate sanitizer solution often.

XXIII. Temporary Traffic Control

A. Basic Requirements

Temporary traffic control (TTC) is an important aspect of surveying work and applies anytime work is being performed near or in a road right-of-way (ROW), including a low traffic volume residential street or county road. The primary function of TTC is to provide for the reasonably safe and effective movement of road users through or around TTC zones while reasonably protecting road users, workers and responders to traffic incidents, and equipment. Worker safety is of equal importance.

When performing surveys adjacent to or within a major collector or a State highway ROW traffic control becomes a more serious matter due to higher speeds and greater volumes of traffic.

It should also be noted that liability exists when the responsibility of controlling traffic is assumed.

The respective entity should be contacted prior to implementing any traffic control setup that involves work near or in traffic lanes. Responsibility for enforcement of the requirements is the respective city or county's, though for the safety of workers and the traveling public, implementation of appropriate traffic control is incumbent on the survey company and survey crew performing work within a public ROW.

In a State ROW, a traffic control setup should only be implemented when proper notification is given to the appropriate Texas Department of Transportation (TxDOT) District Office and to the local TxDOT Area Engineer's Office which has direct responsibility for all activities within all State ROW within their respective area of responsibility. Traffic control requirements, devices, and implementation plans shall be discussed with the TxDOT staff prior to implementing any traffic control setup. Strict adherence to all requirements is a must and all plans must be

followed to ensure the safety of all who are working within a State ROW and especially near the road shoulders or surface or on the road surface.

B. Minimum

The minimum traffic control devices for a survey crew should be portable "SURVEY CREW AHEAD" signs and safety cones. Use of safety cones should be considered any time a survey vehicle is stopped, and workers are accessing or replacing survey instruments and equipment.

Traffic cones should also be used around a survey instrument in a setup that can be accessed by onlookers, pedestrians, or moving equipment.

Signs should be placed in advance of the work area to alert drivers that work is taking place. The distance that a sign or signs should be placed in advance of the work zone is a function of several variables that include the speed limit of the roadway, cross-streets, and terrain conditions.

C. Line of Sight Adjustments Due to Curves, Hilly Terrain, and Other Conditions

Even with standard recommended plans, consideration shall always be given to curves, terrain, weather, and other conditions unique to the location. Additional TTC devices, lengthening of a layout or other adjustments may be called for to increase the safety of workers and the road users.

D. Texas Manual on Uniform Traffic Control Devices

The Texas Manual of Uniform Traffic Control Devices (TMUTCD) is the Federal Highway Administration (FHWA) approved State of Texas version of the federal Manual of Uniform Traffic Control Devices and is required for every state. Its use is required for selecting all appropriate traffic control signs and devices as well as basic traffic control setups. The TMUTCD is available through the Texas Department of Transportation web site at: *https://www.txdot.gov/government/enforcement/signage/tmutcd.html*

The devices and plans depicted in the TMUTCD are standard traffic control signage and the traffic control plans depicted should be considered as typical and a minimum requirement when performing surveying work or setting up instruments within a public road or street right-of-way (ROW), or within a private thoroughfare such as a private street, shopping center or office complex that is accessible to the public. Conditions of a specific work location should be considered and appropriate modifications to the layout such as additional traffic control signs, modified minimum sign spacing, etc., should always be applied. Additionally, any jurisdiction (town, city, county, etc.) in which work is being performed within a public ROW has the right to require additional devices and measures dependent upon traffic conditions and issues at the site of proposed work.

XXIV. Utilities – Above Ground and Underground

It is imperative that survey crew personnel always be alert for the presence of above ground utilities and even more importantly for the possibility of underground

utilities, even when performing normal digging, searching, and probing for property corners. Laws apply to excavation in the vicinity of underground utility locations. By definition in the Texas Administrative Code:

Excavate - Movement of earth by any means.
Excavator - A person that engages in or is preparing to engage in the movement of earth.

Damage to an underground utility from a dig in can not only present a danger, but the disruption of service can also cause an inconvenience as well as a monetary fine to the party causing the damage.

Above ground and underground utilities consist of many types: electric; telecom-telephone, cable TV, internet; water and sewer; oil and gas- natural gas, oil, chemical, gasoline, to name a few. Pipelines can also be low pressure (under 200 psi) or high pressure (200-1500 psi).

Above ground electric power lines present an electrocution hazard if touched by a conductor such as an extended level rod or a prism or GPS unit on a rod. With respect to high voltage lines, even encroaching within a 10' radius can be an electrocution hazard.

Underground utilities present a dig in hazard even when conducting normal surveying activities in the vicinity of an underground utility. A dig in of a pipeline carrying hazardous or flammable product can cause a fire or explosion. A dig in of a buried electrical line can cause shock or electrocution hazard.

Sometimes underground utilities are marked and may be apparent. More often, they are not. Even if signs indicating the approximate location of the respective utility are present the sign or marker cannot be assumed as the exact location of the underground utility.

Always call Texas811 for a locate before conducting and excavation as defined by applicable law.

XXV. Texas One Call – Texas811

It is imperative to call or contact Texas811 prior to excavating, digging, placing property boundary monuments, survey control monuments, marker posts or any other disturbance below the surface of the earth.

First, mark the "dig" location(s) as required using white paint before calling 811.

Contacting Texas811:

- Call 811 or use the Online Ticket Entry System, The Portal, at *https://txgc.texas811.org/geocall/portal.*
- Provide the required information.
- Wait 48 hours (2 business days).
- If no response, contact Texas811 again.

An Excavator is responsible for:

- Requesting the one-call notification; it is not your client's responsibility.
- Immediately reporting the damage or exposure of any underground utility to the one-call center. If a gas line has been hit or damaged, you must also call 911.
- Knowing and following the tolerance zones for digging around the locators' markers.
- Providing your client with the appropriate ticket or notification number prior to beginning work.
- Requesting a line locate at least two business days, excluding weekends and holidays, before beginning work.
- Providing accurate driving directions with distances and cardinal headings (North, South, East, West) when working in rural areas.
- Confirming the response of the notified utilities before proceeding with any excavation.
- Notifying the one-call center after determining that one or more utilities have not responded to a locate request.
- Looking for utility signage at your work site and obeying instructions on those signs.
- Following safe digging practices and maintaining equipment condition to ensure safety standards.
- Look over the Pre-Excavation Checklist.
- Read the Texas811 Excavator Guide with Ch 18 and Ch 251.

Necessary Information to Keep - After calling Texas811 you will receive to keep:

- A ticket number, your proof that you contacted Texas 811 and requested a locate.
- An email copy of the information and the list of utilities that have facilities in your area.
- An identical ticket from the other contact center in Texas listing utilities they notified.

Other Utility Companies:

A. Above Ground and Below Ground

- Electric Lines
- Cable TV and Telephone

B. Underground

1. Pipelines

   Pipelines pose a special concern to surveying operations since they may contain a variety of hazardous chemicals, gas, petroleum products, other substances, or water. Pipelines can be low pressure or high pressure, up to or exceeding 1400 psi. Whether the product is hazardous or not, the high pressure alone can be deadly. If damaged by surveying operations, the damage to the

pipeline can be expensive for the owner/operator and the cost of repair and fine to the surveyor can be high.

Pipelines include:

a. Crude Oil
b. Chemical
c. Natural Gas
d. Other Petroleum Products
e. Railroad Commission of Texas has jurisdiction over some pipelines.

2. Telephone – Conventional and Fiber Optic

Telephone lines are normally marked with signs that indicate the owner and contact information or telephone number. Telephone boxes/pedestals are often visible. Some telephone cables are enclosed in a form of conduit. Some lesser capacity telephone cables are not encased. Be especially alert to the fact that sometimes, buried telephone cables are not in an exact straight line, and that signs and markers may be offset a varying distance from the actual cable or conduit.

Some smaller, rural telephone companies may not be a part of the Texas 811 system and will need to be contacted separately.

3. Public and Private Water Lines

Water lines are typically owned, operated, and maintained by a local entity such as a city or municipal utility district. Some which may not be part of the Texas811 notification network. Others, especially in suburban and rural areas may also be owned and operated by private companies.

Pay special attention to markers, if any, apparent trench lines or other signs indicating the presence of a buried water line or other utility. Be sure to contact the city, water company, Municipal Utility District, or any other utility owner/operator separately and in addition to the 811 notification.

4. Sanitary Sewer Lines

Sanitary sewer main lines are typically owned and operated by cities, towns, or utility districts. Service lines, from the main line to a private structure are likely owned in part by the above and in part by the landowner. Contact the local city, town, or utility district for information on these lines.

5. American Public Works Association (APWA) Uniform Color Code for Marking Underground Utility Lines:

The following are the standard APWA colors for marking the respective utilities or proposed excavation. Marks may be painted on the ground, or wire flags or stakes with flagging may be placed over the respective utility or excavation location.

WHITE - Proposed Excavation

PINK - Temporary Survey Markings

RED - Electric Power Lines, Cables, Conduit and Lighting Cables

YELLOW - Gas, Oil, Steam, Petroleum or Gaseous Materials

ORANGE - Communication, Alarm or Signal Lines, Cables or Conduit

BLUE - Potable Water

PURPLE - Irrigation and Slurry Lines, Reclaimed Water

GREEN - Sewers and Drain Lines

XXVI. Vehicles and Survey Trucks

A. Driving

See Section VIII on Driving.

B. Job Site Safe Parking

Always consider parking in a safe location and in the safest manner possible when on a job site.

- Safe access for the survey crew while removing instruments, tools, and supplies, and while replacing these items upon completion of work.
- Job site construction equipment movement if within a work zone.
- Traffic, especially when parking within a high volume, high speed road ROW.
- Terrain, especially with respect to whether the parking location is flat, on a slope, on the shoulder of a road, and so on.
- Remember that when parking a vehicle, Texas law requires setting the vehicle's parking brake and turning the wheels up slope to prevent rolling.

C. Vehicle Maintenance

Vehicle maintenance can be a matter of safety. Ensuring that a vehicle is in good mechanical condition and that a vehicle has appropriate tires, in good condition and properly inflated could be a matter of life or death while that vehicle is in motion.

D. If an Incident Occurs (If you are involved in a wreck situation)

1. Incident Checklist
   a. Protect the incident scene to prevent additional incidents.
   b. Call 911 if there are injuries.
   c. Call appropriate law enforcement.
   d. Give reasonable aid to the injured, law enforcement or other authorized persons at the scene.
   e. Allow use of your first aid kit or fire extinguisher by competent persons.

f. Call an emergency response services (911), doctor or towing service if needed.
g. Contact your company representative- supervisor, lead worker, etc.
h. Do not give statements to anyone other than law enforcement, your company or company authorized representatives.
i. Get Details. Get driver information, insurance company information, their cell phone number.

XXVII. Water Operations

1. When surveyors are exposed to the hazard of drowning, they shall be supplied with U.S. Coast Guard - approved life jackets, ring buoys with lifeline, lifeline, or any combination of the three. Make certain that any life jackets, rings, or line used is sound and in good condition. Replace any that are not in acceptable condition and dispose of unacceptable devices.

2. In Boats
   a. Wear a Coast Guard approved life jacket if the water is over waist deep.
   b. Wear appropriate anti-slip footwear.
   c. Use only boats that are well maintained and in safe condition.
   d. Follow the common rules of boating safety.
   e. Only use boat operators who have been trained in boating safety.
   f. Work with a buddy.
   g. Do not overload a boat.

3. In the Water
   a. Do not wade barefoot. Wear rubber boots to protect against cuts, cold and water.
   b. Work with a buddy.
   c. Wear a U.S. Coast Guard approved life jacket. if the water is over knee high or if working in current or surf waters.
   d. In still water:
      - Limit wading to waist-deep locations. Wear chest waders with a waist belt.
      - Probe with a pole for holes and soft ground before proceeding.
   e. Wading in moving water - Do not wade if:
      - Any doubt exists as to safety.
      - The water is more than mid-thigh high.
      - The product of stream velocity in feet per second (f.p.s.) and stream depth in feet equals ten or more. Example: 3 fps x 4’ depth = 12.
      - Footing is difficult to retain because of any or a combination of stream velocity, streambed roughness or slipperiness, unless a tautly stretched lifeline is used as a handrail.

XXVIII. Definitions and Terminology

Accident - an unforeseen and unplanned event or circumstance that sometimes has inconvenient or undesirable consequences, other times being inconsequential. The event may or may not have unrecognized or unaddressed risks contributing to its cause.

Allergic Reaction - a sensitivity to something that was eaten, inhaled, or touched. Your body interprets an allergen, something your allergic to, as foreign or harmful, and it attacks it as a form of protection.

ANSI - acronym for the American National Standards Institute; a private, non-profit organization that administers and coordinates the U.S. voluntary standards and conformity assessment system.

ASTM - now ASTM International, formerly the American Standards and Testing and Materials; one of the world's largest international standards developing organizations.

CPR/Cardiopulmonary Resuscitation - “an emergency lifesaving procedure performed when the heart stops beating.”

Defensive Driving - "driving to save lives, time, and money, in spite of the conditions around you and the actions of others," as defined by Safe Practices for Motor Vehicle Operations, ANSI/ASSE Z15.1.

Distracted Driving - any activity that diverts attention from driving, including talking or texting on your phone, eating and drinking, talking to people in your vehicle, fiddling with the entertainment or navigation system, or daydreaming. Anything that takes your attention away from the task of safe driving.

Emergency - a situation that endangers life, health, or property or a situation in which the public need for uninterrupted service and immediate re-establishment of service if services are interrupted compels immediate action.

Excavate - movement of earth by any means.

Excavator - a person that engages in or is preparing to engage in the movement of earth.

First Aid - emergency care or treatment given to an ill or injured person before regular medical aid can be obtained.

GPS - Global Positioning System. A satellite-based radionavigation system owned by the United States Space Force or by other nations around the world.

Hand Tool - any tool held in the hand and operated without other power.

Hazardous - dangerous, risky, unsafe, unpredictable.

Heat Illness - heat illness, heat exhaustion, heat stroke

Horse Play - rough, boisterous play, foolish behavior.

Hydration - the process of causing something to absorb water. Drinking adequate water.

Hyperthermia - the condition of having a body temperature greatly above normal.

Hypothermia - the condition of having an abnormally low body temperature, typically one that is dangerously low.

Incident - an event or occurrence; one that may or may not be avoidable; unintended event that is often unpleasant.

ISEA - acronym for the International Safety Equipment Association; an association for safety equipment and technologies.

Near Miss - a narrowly avoided collision or other accident. A "near hit" or a close call.

Personal Protective Equipment (PPE) - equipment or clothing worn that is designed to keep people safe and to minimize exposure to workplace hazards that cause serious injuries and illnesses. This includes items such as safety glasses or goggles, safety-toe boots or shoes, gloves, headgear, hearing protection, high-visibility clothing (vests, safety shirts), respiratory protection and more.

OSHA - The Occupational Safety and Health Administration. A regulatory agency of the United States Department of Labor whose mission is to ensure safe and healthful working conditions for working men and women by setting and enforcing standards and by providing training, outreach, education, and assistance.

Pinch Point - Per OSHA standard "1910.211(d)(44), "Pinch point" means any point other than the point of operation at which it is possible for a part of the body to be caught between the moving parts of a press or auxiliary equipment, or between moving and stationary parts of a press or auxiliary equipment or between the material and moving part or parts of the press or auxiliary equipment." Can also be where two objects or parts come together such as when a pickup truck tailgate closes, or a toolbox lid is closed.

Poisonous - causing or capable of causing death or illness if taken into the body.

PSI - pressure in pounds per square inch.

Recommendation - suggestion or proposal given as the best course of action.

Requirement - something that is mandatory or compulsory.

Risk - A probability or threat of damage, injury, liability, loss, or any other negative occurrence that is caused by external or internal vulnerabilities, and that may be avoided through preemptive action.

Texas Manual on Uniform Traffic Control Devices - (TMUTCD). A manual adopted by TxDOT in compliance with Section 544.001 of the Texas Transportation Code. The manual contains specifications for a uniform system of traffic control devices for use on streets, roads, and highways within the state.

Texas One Call 811 - Texas811 is a non-profit organization that works with over 1,600 utility and municipal members in the state of Texas. The organization runs a complex call center that connects surveyors and excavators to local utility providers to ensure underground utility lines are properly identified before construction.

Temporary Traffic Control - TTC- The primary function of TTC is to provide for the reasonably safe and effective movement of road users through or around TTC zones while reasonably protecting road users, workers, responders to traffic incidents, and equipment.

Venomous - capable of secreting venom; capable of injecting venom by means of a bite or sting.

XXIX. Resources

A. Websites and Links

American Heart Association: *https://cpr.heart.org/en/resources/what-is-cpr*

Medical Information, First Aid

First Aid, Mayo Clinic: *https://www.mayoclinic.org/first-aid*

First Aid Steps, Perform First Aid, Red Cross:

https://www.redcross.org/take-a-class/first-aid/performing-first-aid/first-aid-steps

WebMD.com: *https://www.webmd.com/first-aid*

Operation Game Thief (OGT): *https://tpwd.texas.gov/warden/connect/ogt*

OSHA, Occupational Safety and Health Administration: *https://www.osha.gov*

Texas 811: *https://txgc.texas811.org/geocall/portal*

Texas Manual on Uniform Traffic Control (TMUTCD):

https://www.txdot.gov/government/enforcement/signage/tmutcd.html

Texas Parks and Wildlife Department (TPWD): *https://tpwd.texas.gov/*

Texas Poison Center Network: https://www.poisoncontrol.org/

Telephone: 800-222-1222